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AQA

Please write clearly in block capitals.

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I declare this is my own work.

Functional Skills Level 2 MATHEMATICS

Paper 2 Calculator

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:
- a calculator
- mathematical instruments.



Instructions
- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

For Examiner's Use	
Question	Mark
1–7	
8	
9	
10	
11	
TOTAL	

OFFICIAL MFS MODEL SOLUTION · BENCHMARK EXEMPLARY PASS

Full Marks / Exemplary (100% Pass · Model Solutions)

· Assessment: March 2022 - Paper 2

· Subject: AQA Functional Skills Maths Level 2

· Complete accredited step-by-step models and tutor guidance

· Exam Board: AQA (Ofqual Regulated Reform Standard)

Verified by Tutor Abdul (Senior Assessor) · MasterFunctionalSkills.co.uk

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Section A

Answer **all** questions in the spaces provided.

1 Circle the **smallest** number.

[1 mark]

1 −5 −3 0

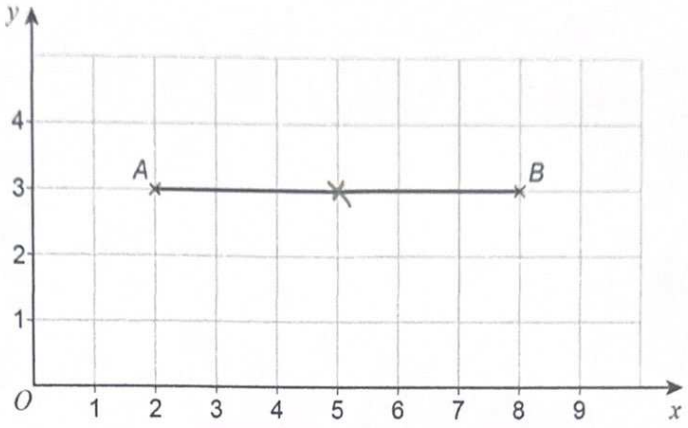
2 Complete the table to show equivalent fractions, decimals and percentages.

[3 marks]

Fraction	Decimal	Percentage
$\frac{17}{20}$	0.85	85%
$\frac{1}{25}$	0.04	4%
$\frac{59}{100}$	0.59	59%

Do not write outside the box

3 $A(2, 3)$ and $B(8, 3)$ are points on a grid.



Work out the coordinates of the **midpoint** of line AB.

[1 mark]

Answer (5 , 3)

4 Here are six numbers.

7 10 13 7 15 8

Work out the median.

[2 marks]

7, 7, 8, 10, 13, 15.

$\frac{8+10}{2} = 9$.

Answer 9

Do not write outside the box

5 Express 76 as a percentage of 200

[1 mark]

$\frac{76}{200} \times 100 = 38\%$

Answer 38 %

6 Share 126 in the ratio 3 : 11

[2 marks]

$3+11=14. \quad 126 \div 14 = 9.$

$3 \times 9 : 11 \times 9$

$= 27 : 99$

Answer 27 and 99

Do not write outside the box

7 Which is closer to 9.5

9.073 or 9.916 ?

You **must** show your working.

[2 marks]

$9.5 - 9.073 = 0.427$.

$9.916 - 9.5 = 0.416$. ←

9.916 is closer.

Answer 9.916.

12

Turn over for the next section

Do not write outside the box

Section B

Answer **all** questions in the spaces provided.

8 Pets

8 (a) Miya manages a pet shop.

She reduces the price of all fish tanks by 17%

She puts this sticker on one of the tanks.

Reduced by 17%
Was £195
Now £167.95

Has Miya calculated the new price correctly?

You **must** show your working.

[3 marks]

$195 \times (1 - 0.17) = 195 \times 0.83$

$= £161.85.$

$\neq £167.95.$

Miya is wrong.

Do not write outside the box

8 (b) Alex makes food for his dogs.

He uses 2 **pounds** of mince to make 12 portions of food.

He has 1.5 **kilograms** of mince.

Is this enough to make 21 portions of food?

Use 1 kg = 2.2 pounds

You **must** show your working.

[3 marks]

1.5kg = 2.2 × 1.5 = 3.3 pounds.

2 ÷ 12 = 1/6 of a pound per portion.

1/6 × 21 = 3.5 pounds for 21 portions.

So Alex does not have enough for 21 portions.

Question 8 continues on the next page

Do not write outside the box

8 (c) Sophie has a hutch for her pet rabbit.

The floor area of the hutch is $4.46\,m^2$

Sophie builds a new hutch with the floor in the shape of a regular octagon.

Area of a regular octagon $= 2s^2(1+\sqrt{2})$ where s is the side length

The side length of the floor of the new hutch is 1.1 m

How much bigger is the floor area of the new hutch than the floor area of the old hutch?

[3 marks]

$S = 1.1$

Area $= 2 \times 1.1^2 \times (1+\sqrt{2}) = 5.84239...$

$5.8424 - 4.46 = 1.3824$

Answer 1.3824 m^2

9

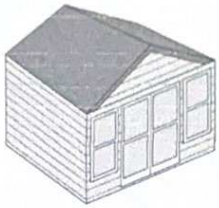
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9 Summer house

Lisa is buying a summer house for her garden.

9 (a)

Summer house	£5964
Delivery charge	£2.25 per mile



Payment plan

Deposit of $\frac{2}{7}$ of the total price including delivery

then 12 equal monthly payments

Lisa buys the summer house, with a delivery charge for 84 miles.

She uses the payment plan.

Work out the deposit and the amount of each monthly payment.

[6 marks]

Delivery cost = 84 × 2·25 = £189.

Total cost = 5964 + 189 = £6153.

Deposit: 6153 × 2/7 = £1758.

Left to pay = 6153 − 1758 = £4395.

Monthly payment: 4395 ÷ 12 = £366·25.

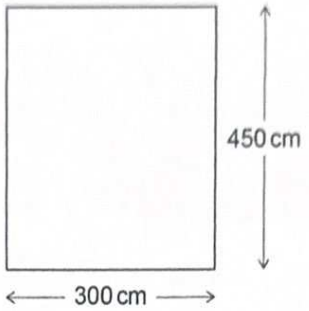
Deposit £ 1758.

Monthly payment £ 366·25

Turn over ▶

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9 (b) The base of the summer house has dimensions 300 cm by 450 cm



Lisa's garden is drawn to a scale of 1 to 100 on a centimetre grid.

There is a wall at the bottom of the garden and a flower bed down one side.

Lisa wants to put the summer house

at least 400 cm from the wall

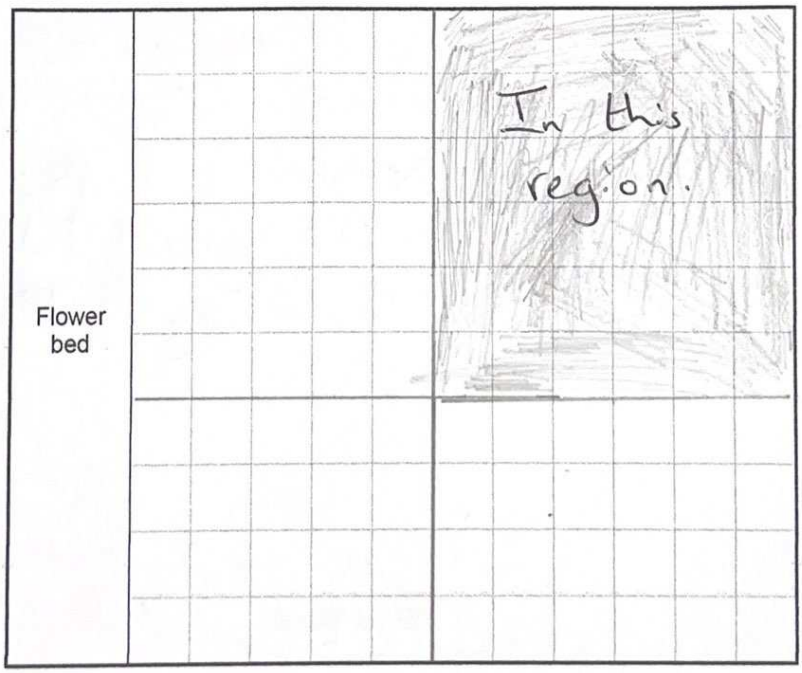
and

at least 500 cm from the flower bed.

On the centimetre grid, show where Lisa can put the summer house.

[3 marks]

Scale: 1 to 100



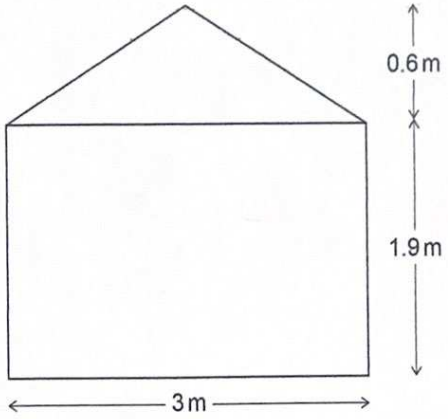
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9 (c) Lisa is buying a heater for the summer house.

To work out the correct heater size she uses the formula

Minimum heater size (kW) = area of the front (m^2) × length (m) × 0.058

Here is the front view of the summer house.



Not drawn accurately

The length of the summer house is 4.5 m

Lisa can buy a 1 kW heater or a 2 kW heater.

Which heater should she buy?

You **must** show your working.

[5 marks]

Area of rectangle = $3 \times 1.9 = 5.7 m^2$

Area of triangle = $\frac{1}{2} \times 3 \times 0.6 = 0.9 m^2$.

Total area of front: $6.6 m^2$.

Minimum heater size = $6.6 \times 4.5 \times 0.058$

$= 1.7226$ kW.

She should then buy the 2kW heater.

Answer 2kW.

14

Do not write outside the box

10 Earnings and saving

10 (a) Jamal works for a company.

His normal rate of pay is £9.50 per hour.

His overtime rate of pay is 1.5 times his normal rate.

One week, Jamal works

38 hours at his normal rate

and

6 hours overtime.

From his earnings, the company deducts

Income Tax at 20% on his earnings over £242

National Insurance at 12% on his earnings over £184

Work out Jamal's net pay that week, after Income Tax and National Insurance are deducted.

[7 marks]

Jamal's overtime pay = $9.50 \times 1.5 = £14.25/hr$.

Total weeks pay = $38 \times 9.50 + 6 \times 14.25$

$= £446.50$.

Taxable: $446.5 - 242 = £204.50$

Tax: $204.50 \times 0.2 = £40.9$

National Insurance able: $446.5 - 184 = £262.50$

N.I: $262.50 \times 0.12 = £31.50$.

Net pay = $446.5 - 40.9 - 31.5$

$= £374.10$.

Answer £ 374.10.

Do not write outside the box

10 (b) Jamal decides to put £2500 in a savings account for 3 years.

He compares the interest paid by two banks.

Bank A	Bank B
1.4% compound interest per year	Year 1 – no interest added Year 2 – £49.15 interest added Year 3 – £56.30 interest added

Which bank should Jamal choose, A or B?

You **must** show your working.

[5 marks]

Bank A: 2500×1.014^3

$= £2606.48.$

Bank B: $2500 + 49.15 + 56.30$

$= £2605.45.$

Therefore bank A will net Jamal more interest.

Answer Bank A.

12

Do not write outside the box

11 Soft Play Centre

11 (a) At a soft play centre one morning,

there are 6 staff on duty

the ratio of staff to children is 2 : 17

35 of the children are aged under 7

How many of the children are aged 7 or over?

[4 marks]

Staff : Children

$\times 3$ (2 : 17) $\times 3$

6 : 51

There are 51 children total.

51 − 35 = 16 children aged 7 and over.

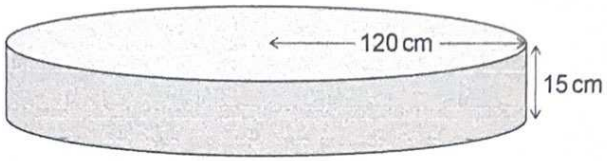
Answer 16

Do not write outside the box

11 (b) A new sand pit is built at the soft play centre.

The sand pit is in the shape of a cylinder of radius 120 cm

The sand pit will be filled with play sand to a depth of 15 cm



Play sand is sold in 50-litre bags and each bag costs £9.97

1 litre = 1000 cm^3

How much will it cost to buy enough bags of sand to fill the sand pit to a depth of 15 cm?

[6 marks]

Area of the cylinder is $\pi \times 120^2 \times 15$

$= 678584.01 cm^3$

$= 678.584 L.$

Bags needed $= 678.584 \div 50$

$= 13.57...$ bags.

Therefore 14 bags are needed.

Cost $= 14 \times 9.97 = £139.58.$

Answer £ 139.58.

Do not write outside the box

11 (c) The table shows the ages of children at the centre one day.

Age	Number of children
3 and under	8
4–6	15
7–9	13
10 and over	4

One of the children is chosen at random to win a free soft play visit.

The manager says,

"The probability that the child is aged 7–9 is **more than** $\frac{3}{10}$"

Is the manager correct?

You **must** show your working.

[3 marks]

Total Children = $8+15+13+4=40$.

Probability $=\frac{13}{40}$.

$\frac{3}{10}=\frac{12}{40}<\frac{13}{40}$.

So yes, the manager is correct.

13

END OF QUESTIONS

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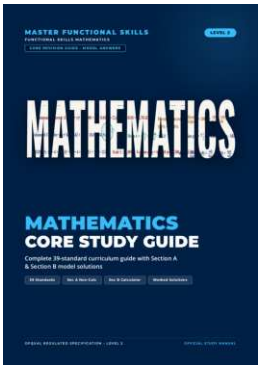
Question number	Additional page, if required. Write the question numbers in the left-hand margin.

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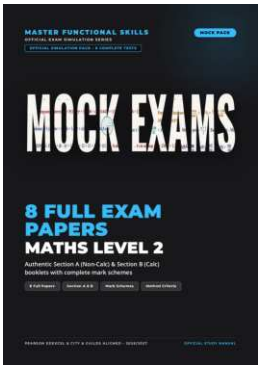


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